Salah Boubnider University

Faculty of Medicine of Constantine

Cytology Courses

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Lecture 11

The Mitochondrion
LA Mitochondrie

Generally distributed throughout the hyaloplasm, the mitochondrion is a closed organelle present in all cells of eukaryotic organisms. It sometimes exhibits clusters, as is the case in some oocytes, or preferential localization related to specific energy needs. In general, the number and total volume of mitochondria are proportional to the cell volume; in a hepatocyte, for example, approximately 1500 mitochondria occupy 20% of the cell volume.

1. Structure and Distribution of Mitochondria

These highly dynamic organelles appear as more or less elongated granules (Figure 1); globular (0.5 to 1 µm in diameter), or filamentous, up to 10 µm long, organized into a network and composed of tubules that constantly fuse.
Mitochondria possess two membranes: the outer membrane and the inner membrane, which separates the mitochondrial matrix from the intermembrane space. The latter forms invaginations in the mitochondrial matrix called cristae, which are the site of phosphorylation oxidation reactions during which ATP synthase uses the electrochemical gradient of protons generated by the respiratory chain to synthesize ATP.

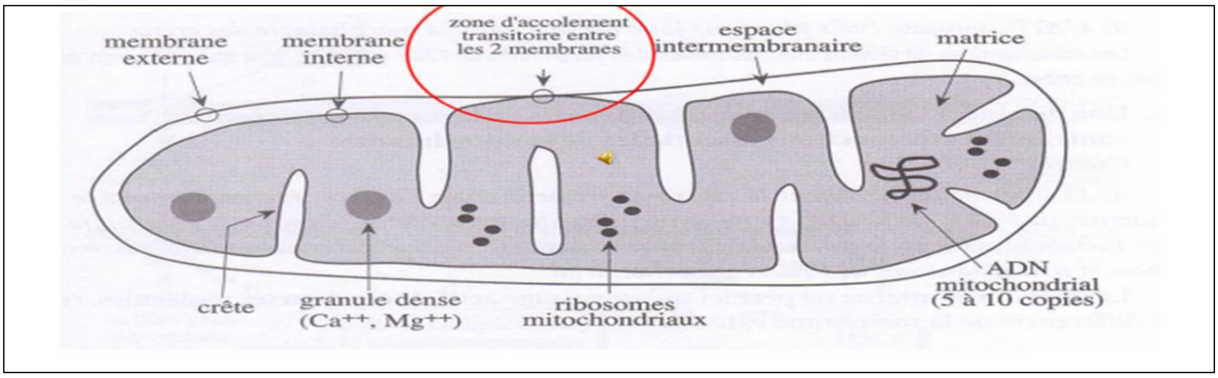

Figure 01. Schematic representation of the ultrastructure of a mitochondrion.

2. The chemical composition of the mitochondrion

Mitochondria consist of an outer membrane and an inner membrane that define the intermembrane space and the mitochondrial matrix.

2.1. The outer membrane: This is a lipid bilayer 5 to 7 nm thick, with a composition similar to that of the plasma membrane but containing more protein. It is very rich in porins and permeable to ions and molecules with a molecular mass of up to six kDa. Several enzymes involved in lipid metabolism and protein transporters represent proteins.

2.2. The intermembrane space: This is a dense space 4 to 7 nm thick, containing H+ protons that play an important role in phosphorylation, cytochrome C molecules, and molecules generally smaller than 10 kDa.

2.3. The inner membrane: This is a lipid bilayer 5 to 6 nm thick, characterized by a very different organization from that of the outer membrane, as it is composed of 80% proteins and 20% lipids. It is very rich in transporters and protein-enzyme complexes, and it contains TIM translocases involved in protein import.

2.4. The mitochondrial matrix: This occupies the inner space and contains mitochondrial ribosomes, circular DNA (mtDNA) molecules, messenger RNA and transfer RNA, dense and irregular granules formed by the accumulation of calcium and magnesium cations, and numerous enzyme systems.

2.5. Mitochondrial DNA: Mitochondria possess their own genome (mtDNA), which is a molecule of maternal origin.

3. Physiological Roles of Mitochondria

The involvement of mitochondria in numerous cellular functions has made this organelle a central element of cell life.

3.1. Import of Cytosolic Proteins:

The import of cytosolic proteins into the mitochondrion (Figure 2) occurs in several steps:

1. A cytosolic Hsp70 protein triggers the unwinding of the protein to be imported, allowing it to cross the two TOM and TIM complexes.

2. The targeting sequence is recognized by a protein receptor on the outer membrane.

3. The receptor triggers the translocation of the protein through the TOM protein complex on the outer membrane.

4. Translocation of the protein continues through a second import complex on the inner membrane, TIM.

5. Proteins that have entered the matrix undergo maturation:
A. Cleavage of the targeting sequence by proteases.
B. The imported protein takes its final conformation through the intervention of Hsp60 and Hsp10 proteins.

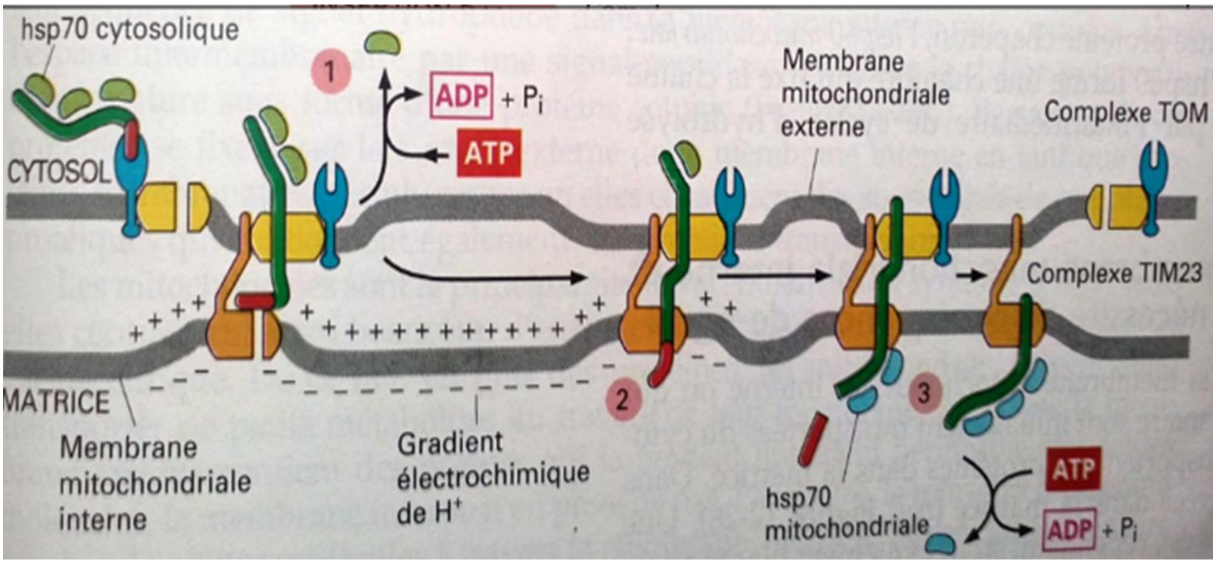

Figure 2: Importation of synthesized proteins into the cytosol under the control of the nuclear genome.

3.2. Steroid hormone synthesis:

Endocrine glandular cells, which secrete steroid hormones (estrogens, progesterone, testosterone, corticosterone, etc.), possess mitochondria that play a role in their synthesis. Mitochondria cooperate with the smooth endoplasmic reticulum (SER) in the synthesis of these hormones (Figure 3).

The inner mitochondrial membrane carries a family of transmembrane enzymes, the cytochrome P450s, whose active site is bathed in the matrix. Other enzymes belonging to this family are anchored in the SER membrane, whose active site is bathed in the cytosol.

These enzymes are hydroxylases, which use O2 and electrons from NADPH (matrix or cytosolic NADPH from pentose phosphate groups) to hydroxylate cholesterol molecules into pregnenolone in the mitochondrial matrix.

Pregnenolone leaves the mitochondria and reaches the cytosolic side of the smooth endoplasmic reticulum (SER) membrane.

SER cytochromes P450 synthesize two types of pregnenolone derivatives:

1. Sex hormones (sex steroids)
2. Intermediate metabolites that return to the mitochondrial matrix, where other cytochromes P450 in the inner membrane then use them for the synthesis of cortisol and aldosterone.

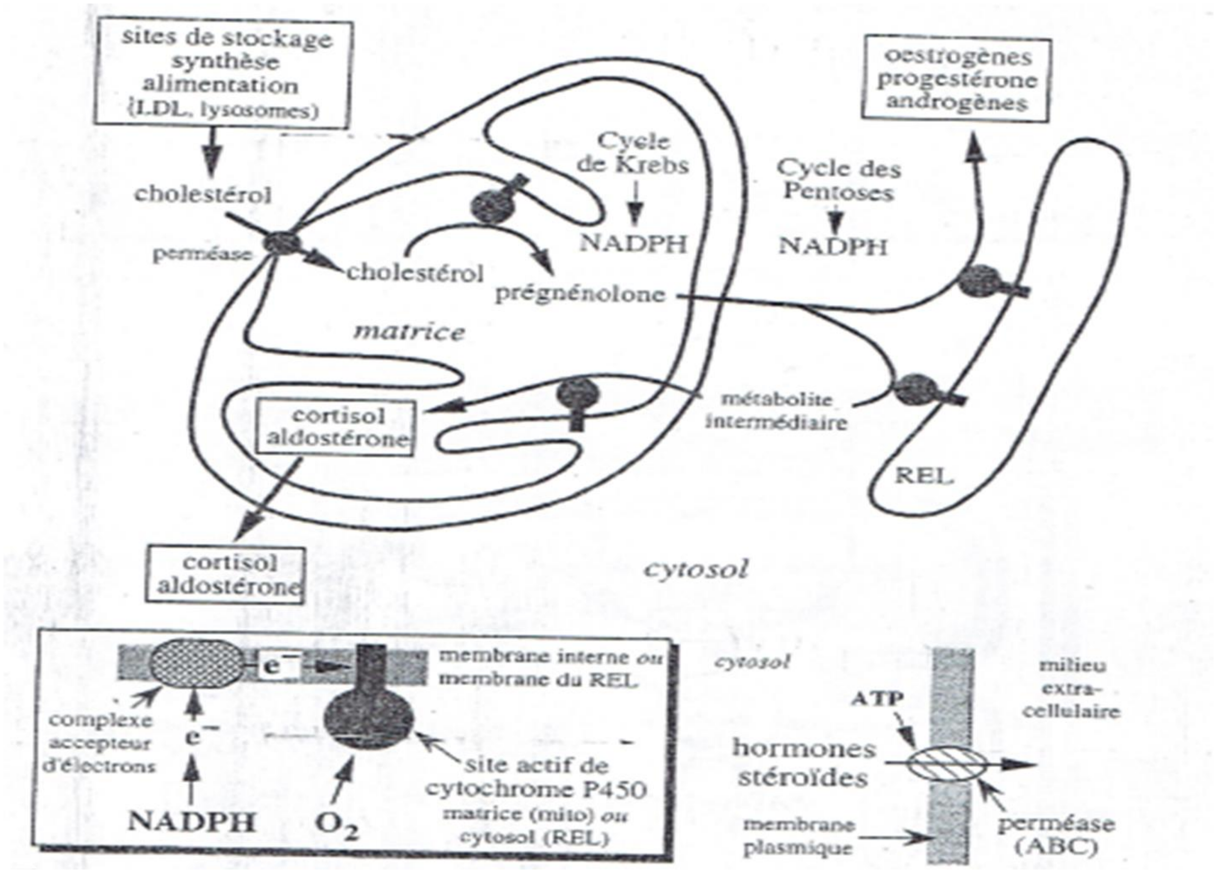

Figure 03: Synthesis and export of steroid hormones

3.3. Protein Synthesis under the Control of Mitochondrial Genome:

Mitochondrial DNA contains 37 genes that code for 13 proteins, 22 transfer RNAs, and 2 ribosomal RNAs.

All the encoded proteins are subunits of enzyme complexes of the oxidative phosphorylation system.

3.4. Calcium Storage:

Mitochondria, along with the smooth endoplasmic reticulum, are the main reservoir of calcium. They are capable of capturing calcium, storing it in the matrix, and then releasing it into the cytosol via ion channels in the inner membrane and sodium/calcium exchangers.